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Application of baited remote underwater video surveys to quantify spatial distribution of elasmobranchs at an ecosystem scale

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ABSTRACT

Understanding the distribution of mobile species across broad spatial scales and a variety of habitat types is essential in developing sound conservation strategies. The applicability of baited remote under-water video station (BRUVS) to determine the distribution of elasmobranchs at broad spatial scales (12° latitude) and across a range of habitat types was assessed. Shark-like batoids within the Great Barrier Reef Marine Park (GBRMP) were chosen as a case study, given the conservation concern surrounding these species, limited knowledge of their spatial ecology and their exposure to fishing. *Rhynchobatus* spp. (88.9%), were most frequently observed followed by *Rhina ancylostoma* (7.1%) and *Glaucostegus typus* (3.9%). Shark-like batoids were observed across a wide depth range (10.4–87.5 m), showing highest affinity for 30–40 m. The main predictors of *Rhynchobatus* spp. presence in the GBRMP were location, substrate type and season with individuals more commonly observed during winter and autumn in regions with substrates dominated by mud or gravel. There was no difference in the proportion of BRUVS in which shark-like batoids were observed between fished and unfished marine park zones. Surveys with BRUVS offer a standardized, non-extractive technique for quantifying the spatial distribution of mobile species that are difficult to sample using conventional techniques across broad spatial scales. Information generated using BRUVS may be useful in helping design improved conservation strategies for rare or threatened elasmobranchs.
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1. Introduction

Understanding how a species is distributed within an ecosystem is important to conservation and management planning (Brooks et al., 2011; Colton and Swearer, 2010). Quantitative information describing species distribution allows assessment of overlap with threats such as fishing, pollution and habitat loss. Management responses are likely to be more effective when explicit distributional information for a target species is available. Mounting pressure from fishing (Bonfil, 1994), habitat degradation (Jennings and Kaiser, 1998; Jennings et al., 2008), pollution (Gelsleichter et al., 2005) and climate change (Chin et al., 2010) has seen the decline of some of the world's elasmobranch populations (Dulvy et al., 2008). The mobility of most elasmobranch species presents significant challenges to their assessment and management (Knip et al., 2012). Species ranges may extend across jurisdictional borders, with the critical habitat of species occurring in regions with differing approaches to environmental regulation and management. Understanding species distributions is thus important to rebuilding depleted populations and stabilizing populations of species harvested as primary targets or as a bycatch.

Naturally low abundances, broad geographic distributions, low economic value and poor taxonomic resolution of fisheries records have all contributed to a state of data deficiency for elasmobranch species (Dulvy and Forrest, 2010; Dulvy et al., 2008; Lack and Sant, 2008; Simpfendorfer et al., 2011; Walker, 2004). Knowledge of elasmobranch species distribution and status has traditionally come from fisheries-dependent data in the form of catch and effort statistics derived from fishery logbooks (Baum et al., 2003; Crow et al., 1996; Stobutzki et al., 2002; Walsh et al., 2002) and observer surveys (Burgess et al., 2005; Harry et al., 2011; Marin et al., 1998; Stevens, 1992). However, bias associated with gear selectivity between life history stages and non-random distribution of fishing effort limit the application of these data to describing a species' spatial distribution. Fisheries-independent sampling tends to employ commercially used gears (e.g. gillnet, trawl, seine, long-line) in conjunction with a more robust sample design to survey community diversity and species distribution (Andrew et al., 1997; Simpfendorfer et al., 2002; Walker et al., 2005), but employment of the same gears as fisheries-dependent sampling constrain the data through gear selectivity (McAuley et al., 2007; Simpfendorfer and Unsworth, 1998). Furthermore, sampling tends to be confined to areas where target species are likely to be encountered. Fisheries-independent sampling programs may also be limited by where certain gears can operate. Broad scale exploratory sampling is seldom conducted due to limitations of both time and money. In addition, sampling in defined geographic locations may not provide accurate estimates of species distribution, particularly if normal movements of a species are large relative to the spatial distribution of the survey (Rago, 2004). Targeted sampling to investigate species

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presence in an area may also fail to encompass variations in seasonal migration or foraging patterns (Rago, 2004). Thus traditional sampling programs may not always provide the best data to define species distributions.

Non-extractive survey techniques have been applied to site attached species that are easily enumerated (Harvey et al., 2002), but these techniques may be less effective for more mobile species (Ward-Paige et al., 2010). Diver based underwater visual survey (UVS) has been used to determine the abundance and distribution of elasmobranch species (Castro and Rosa, 2005; Edgar et al., 2004; Robbins et al., 2006). However, short survey times, mobility of target species (Kulbicki, 1998; Sale and Douglas, 1981; Ward-Paige et al., 2010), behavioral responses to divers (Kulbicki, 1998; Watson and Harvey, 2007), depth, and visibility limitations of SCUBA all impede the ability of UVS to fully quantify elasmobranch abundance and distribution. As a result, UVS tends to produce biased population density estimates (Edgar et al., 2004; Sale and Sharp, 1983) particularly when surveying large mobile species such as elasmobranchs (Ward-Paige et al., 2010).

Baited remote under-water video station (BRUVS) offers a standardized, non-extractive technique for assessing species diversity and habitat presence across geographically wide areas, depth ranges and varied habitats (Cappo et al., 2004). Large elasmobranchs have been sighted commonly in BRUVS footage (Brooks et al., 2011; Malcolm et al., 2007; Meekan et al., 2006). The passive nature of this methodology allows for its application in fragile and protected areas, and when dealing with rare and threatened species that may be negatively affected by intrusive gears or capture (Cappo et al., 2001, 2003). Utilization of BRUVS avoids problems of mesh and hook selectivity encountered when using extractive sampling techniques, and some of the biases of behavioral avoidance associated with UVS. However, this technology is not without its own biases, including: attraction to bait, avoidance by prey species due to attraction of predators, reduced effectiveness in low light and high turbidity conditions and the potential to repeatedly count the same individuals in subsequent surveys (Langlois et al., 2006). Despite this, the data generated by BRUVS provides a permanent record allowing impartial, repeatable measurements and enabling standardized data collection (Cappo et al., 2004, 2007). While BRUVS have been used to compare the abundance of sharks between areas (Brooks et al., 2011; Meekan et al., 2006) and investigate prey communities (Gutteridge et al., 2011; Meekan and Cappo, 2004) their utility to determine the distribution of elasmobranchs at broad spatial scales and across a range of habitat types remains untested. The present study examined the distribution of shark-like batoids (families Rhynchobatidae, Rhinobatidae and Pristidae) at broad spatial scales within the Great Barrier Reef World Heritage Area. The specific aims were: (1) examine the utility of BRUVS to determine shark-like batoid distribution, and (2) examine factors that define the distribution of this group of species.

2. Method

2.1. Study species

The giant shovelnose ray *Glaucostegus typus*, whitespotted guitarfish *Rhynchobatus* spp., shark ray *Rhina ancylostoma*, narrow sawfish *Anoxypristis cuspidata*, and green sawfish *Pristis zijsron* are shark-like batoids that have wide geographic distributions and are globally threatened in *International Union for Conservation of Nature* (IUCN) assessments (Compagno et al., 2006b; McAuley and Compagno, 2003a,b; White and McAuley, 2003a,b). Despite their capture in commerical fisheries within the Great Barrier Reef World Heritage Area (Harry et al., 2011), there is limited understanding of how shark-like batoids are distributed within their geographic ranges. *G. typus* is listed as 'Vulnerable' on the IUCN Red List of Threatened Species, based on intensive fishing pressure and population declines in South-East Asia (White and McAuley, 2003b). Although fishing effort in Australia is not as intense as South-East Asia, *G. typus* has been classified as 'high risk' in ecological risk analyses due to distributional overlap with multiple fisheries (notably gill-net and prawn trawl) and low productivity (Salini et al., 2007). The Family Rhynchobatidae consists of three distinct species (*Rhynchobatus australiae*, *Rhynchobatus laevis* and *Rhynchobatus palpebratus*) which have been consistently confused in the literature (Last and Stevens, 2009). Although each of the species within the complex is listed as 'Vulnerable' on IUCN Red List of Threatened Species the extent of individual species decline and range reductions are hard to quantify given the taxonomic confusion (Compagno and Marshall, 2006; McAuley and Compagno, 2003b; White and McAuley, 2003b). Current management strategies within Queensland waters treat the species complex as a single group due to difficulties in identifying species. Thus we have treated all individuals as a group that will herein be referred to as *Rhynchobatus* spp. Very little is known about *R. ancylostoma*. Fisheries data suggests that it is widely distributed throughout the Indo-West Pacific however, it does not appear to be common (McAuley and Compagno, 2003a). *R. ancylostoma* is assessed globally as 'Vulnerable' on the IUCN Red List of Threatened Species due to evidence of population decline, occurrence in multiple fisheries and high value of fins (McAuley and Compagno, 2003a). However, given the paucity of data it is unclear whether this assessment accurately categorizes the threat faced by *R. ancylostoma*. Similar to other sawfish species (Carlson et al., 2007; Cavanagh et al., 2003; Simpfendorfer, 2000) *A. cuspidata* and *P. zijsron* have suffered substantial reductions in abundance with populations now fragmented throughout their range (Compagno et al., 2006a,b). Consequently, both are listed as 'Critically Endangered' on the IUCN Red List of Threatened Species (Compagno et al., 2006a,b).

2.2. Field methods

A series of surveys comprising 2471 deployments of baited remote underwater video stations (BRUVS) were completed in lagoonal and inter-reef waters of the Great Barrier Reef Marine Park (GBRMP) between March 2000 and May 2010 to survey vertebrate communities (Cappo et al., 2004, 2007). The BRUVS configuration consisted of a galvanized, trestle-shaped frame, enclosing a simple camera housing made from PVC pipe with acrylic front and rear ports. Sony Mini-DV HandiCams with wide-angle lens adapters (0.6×) were used in housings. Exposure was set to 'Auto', focus was set to 'Infinity/Manual', and 'Standard Play' mode was selected on the cameras. Detachable bait arms (20 mm plastic conduit) had a 350 mm plastic mesh canister containing 1 kg of crushed sardines (*Sardinops* or *Sardinella* spp.) as bait, lying on the seabed. BRUVS units were deployed with 8 mm polypropylene ropes and polystyrene surface floats bearing a marker flag and were retrieved with a hydraulic pot-hauler. The BRUVS were deployed to provide between 45 and 90 min of video imagery recorded at the seabed (Cappo et al., 2004, 2007).

A total of 2471 BRUVS deployments were conducted between latitudes 10.7° S and 24.2° S, and longitudes 143.38° E and 152.36° E (Fig. 1). BRUVS were deployed in a depth range of 7–115 m, with a mean sampling depth of 36 ± 15.6 m. Some deployments were shortened due to loss of bait, toppling of the BRUVS by currents, or other factors. The mean length of video imagery was 59.81 ± 9.18 min in the overall data set. Only 5% of all deployments exceeded the median of 62.42 min, and 10% were less than 50 min.

2.3. Data analysis

Interrogation of each tape was conducted using a custom interface (BRUVS 1.5.mdb, Australian Institute of Marine Science, 2006) to manage data from field operations and tape reading, to capture the timing of events, and to capture reference images of the seafloor and fish in the field of view. Shark-like batoids observed on tapes were identified to species.

Records were made for each species of shark-like batoid of the maximum number seen together at any one time on the whole tape (Max*N*).

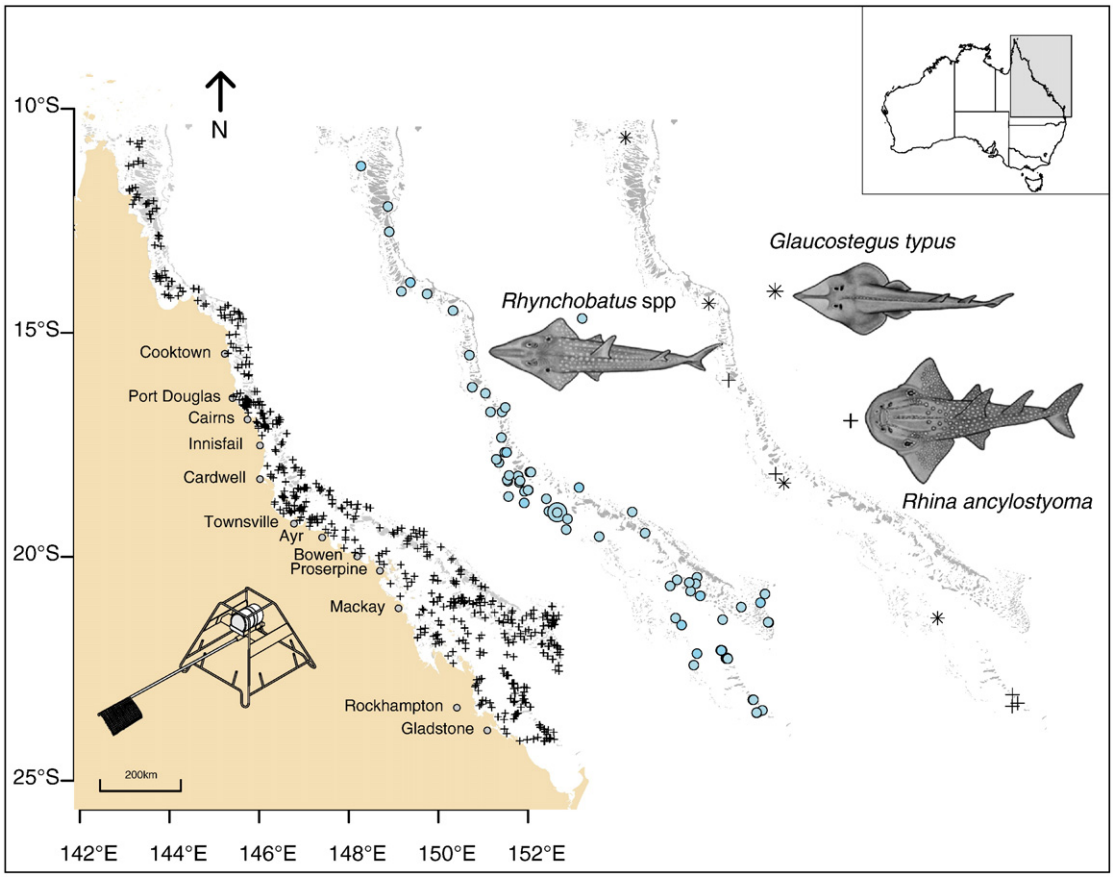

Fig. 1. a) location of all 2471 BRUVS sets within the Great Barrier Reef Marine Park, b) location of sightings of *Rhynchobatus* spp. = ● ✳ = *G. typus*, + = *R. ancylostoma*. All symbols are scaled to a unit of abundance.

The use of Max*N* as an estimator of relative abundance has been reviewed in detail by Cappo et al. (2003).

Depth and latitude in which shark-like batoids were observed on BRUVS were compared to all depths and latitudes at which BRUVS were deployed using Chesson's α (Chesson, 1978):

$$\alpha = (r_i/p_i)/\sum(r_i/p_i)$$

where r_i was the proportion of time a shark-like batoids spent in depth or latitude *i*, and p_i was the proportion of BRUVS at depth/latitude *i*. The value of α can range from 0 (strong avoidance) to 1 (strong affinity), with values > (1/number of categories) indicating electivity and values < (1/number of categories) indicating avoidance.

Chi squared (χ^2) contingency tables were used to compare proportions of *Rhynchobatus* spp. occurring in BRUVS between seasons and fished to no-take areas. Zonation within the Great Barrier Reef Marine Park was changed as part of the representative areas program (RAP) in 2004 (Fernades et al., 2005), which was during the BRUVS sampling period. Therefore pre RAP zonation was used to define fished and no-take area in the present study. Species density and abundance in marine systems typically change with latitude driven by changes in environmental condition (Travers et al., 2012). Community structure is also affected by distance from shore due to species occupying shallow or deeper habitats with preferences driven by variation in abiotic conditions (Travers et al., 2012). Thus depth was used as a categorical variable to assess the effect of distance from shore rather than longitude (Connell and Lincoln-Smith, 1999) as bins of longitude would not allow for meaningful comparisons. Latitude was pooled into six bins to create a categorical variable for analysis. Bins representing 2.5° of latitude were considered the optimal size. *G. typus* and *R. ancylostoma* were excluded from chi squared (χ^2) contingency tables due to limited sample size.

The biological and physical diversity of Great Barrier Reef World Heritage Area (GBRWHA) is divided into 70 bioregions (http://www.soe-townsville.org/data/coastal/marine-bioreg/bioregions_description.pdf). Each bioregion was categorized according to bathymetry, substrate and biology of the area (Fernades et al., 2005). BRUVS was conducted in forty-one of the 70 bioregions within the GBRMP. The substrate type categorical variable (for example fine or coarse sediment, complex rocky, coral, foraminifera/*Halimeda* deposits, gravel, hard substrate seafloor, high carbonate sand, mud, reef, sand, shelly-coarse sediment and terrigenous mud) was taken from the description for each bioregion. There are seven management zonations in the GBRMP designed to regulate activity and conserve biodiversity (Table 1). For the purposes of this analysis zone type (7 factors) and fished/unfished (2 factors) were used. Months of sampling were collated into nominal seasons: summer (December–February), autumn: (March–April), winter (June–August) and spring (September–November) for analysis. The effect of location (latitude/longitude), distance along reef, distance across shelf, substrate type, marine park zone, water depth (continuous) and season on the probability of encountering *Rhynchobatus* spp. was estimated using a logistic generalized linear model (GLM) with a binomial error structure and logit link function. The GLM only considered first order interactions due to limitations in the data. Since the number of individuals observed at individual BRUVS was low, only the presence/absence was considered.

The occurrence (presence) of Rhynchobatus spp. was analyzed using spatial position across and along the GBRMP, and depth, as explanatory

Table 1
Summary of access and use for each of the current marine park zones within the Great Barrier Reef World Heritage Area. Shading indicates zones open to commercial and recreational fishing.

Zone	Fishing	Recreational fishing	Commercial fishing	Access	Tourism
Preservation	No-take	No	No	No	No
Marine national park	No-take	No	No	Yes	Yes
Scientific research	No-take	No	No	No	No
Buffer	Fished	Yes	No	Yes	Yes
Conservation park	Fished	Yes	No	Yes	Yes
General use	Fished	Yes	Yes	Yes	Yes
Habitat protection	Fished	Yes	Yes	Yes	Yes

variables. Introduced by Fabricius and De'ath (2001), the theory behind the use of these "heuristic" spatial predictors was that any sampling region encompassing environmental gradients can be spatially divided into three dimensions in a manner that represents those gradients. Furthermore, it acknowledges that many mechanistic environmental factors can vary along the same gradient, and it is probable that some of these may be important but not measured (or measurable) in a given sampling program. This can lead to spurious inferences about the variables that have been measured. Cross-shelf position was set to 0 on the coast and 1 on the 80 meter isobath on the shelf edge. The along-shelf position was set to 0 on the south-west limit of the GBRMP and 1 on the north-eastern limit. The corners of the polygon formed in this way were 142.530° E, −10.690° S; 144.060° E, −10.680° S at the northern end, and 152.490° E, −25.000° S; 152.900° E, −24.220° S at the southern end.

3. Results

3.1. Shark-like batoid presence

One hundred and twenty individual shark-like batoids were recorded during 2471 BRUVS deployments (Fig. 1, Table 1). *Rhynchobatus* spp. was the most frequently sighted and widely distributed of the shark-like batoids, accounting for 105 (88.3%) of total encounters and occurring across all 12° of latitude. Nine *R. ancylostoma* and five *G. typus* were also present. On two occasions *Rhynchobatus* spp. and *G. typus* were sighted on the same BRUVS. Neither sawfish species known to inhabit the GBRMP (*A.cuspidata*, *P. zijsron*) was observed.

There was no apparent relationship between soak time of the BRUVS and probability of sighting a shark-like batoid. Only two (both *Rhynchobatus* spp.) were sighted after 62.5 min had elapsed.

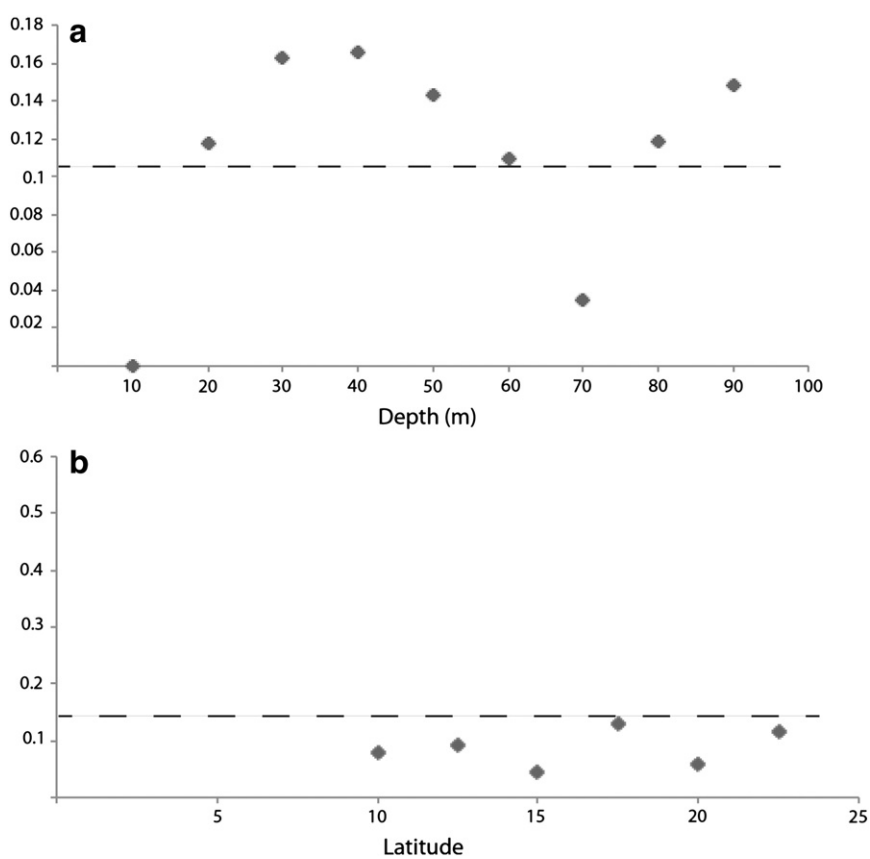

Fig. 2. Affinity of shark-like batoids across (a) depth and (b) latitude ranges at which BRUVS sampling occurred.

The range and means (±standard deviations) of the number of minutes elapsed to first sighting were: for *R. ancylostoma* 9.15–52.5 (30.9 ± 15.3), for *Rhynchobatus* spp. 5.4–80.1 (34.8 ± 16.7), and 5.2–44.7 (31.9 ± 16.6) for *G. typus*.

3.2. Affinity for depth or latitude

Shark-like batoids were present on BRUVS in depths between 10.4 and 87.5 m. Shark-like batoids had the highest affinity for depths from 30 to 40 m (Fig. 2a). Sampling in these depth ranges accounted for 56.7% of total BRUVS effort. Shallow depths (7–20 m; 38.3% of BRUVS) and those greater than 70 m (3.6% of BRUVS) had fewer sightings. Shark-like batoids were present throughout the latitudinal range (Fig. 2b). Fifty five percent of total sampling occurred across five degrees of latitude and accounted for 87.4% of shark-like batoid sightings.

3.3. Effect of season on presence

The proportion of *Rhynchobatus* spp. present between seasons was not significantly different ($\chi^2 = 0.60$, df = 3, p = 0.90), with individuals most commonly recorded during winter (10 of 142 BRUVS; 7.0%) and autumn (33 of 645 BRUVS; 5.1%), while spring (38 of 883 BRUVS; 4.3%) and summer (23 of 801 BRUVS; 2.9%) had lower occurrence.

3.4. Effect of marine park zone on presence

Rhynchobatus spp. were most frequently sighted in habitat protection (37 of 703 BRUVS; 5.0%), marine national park (38 of 814 BRUVS; 4.5%), general use (38 of 753 BRUVS; 4.4%) and conservation park (3 of 74 BRUVS; 3.9%) zones. The number of BRUVS conducted in the port, buffer and preservation zones was too few to draw any conclusions regarding the presence or preference of *Rhynchobatus* spp. for these areas. There was no difference in the proportion of *Rhynchobatus* spp. encountered between zones ($\chi^2 = 0.08$, df = 5, p = 0.993) or between fished and no-take areas ($\chi^2 = 0.004$, df = 1, p = 0.948).

3.5. Effect of bioregion on presence

Shark-like batoids were present in 20 of the 41 bioregions sampled (Table 2), with highest numbers observed in the NN Capricorn Bunker Banks (29 of 285 BRUVS; 10.2%), the NB5 Inner Mid Shelf Lagoon (18 of 188 BRUVS; 9.6%) and NB3 Inner shelf sea grass (12 of 209 BRUVS; 5.7%). Bioregions with the highest number of shark-like batoids present as a proportion of sampling effort were NJ Princess Charlotte Bay Outer Shelf (1 of 6 BRUVS; 16.6%), NK Princess Charlotte Bay (1 of 6 BRUVS; 16.6%), RB1 Far Northern Outer Mid Shelf Reefs (2 of 17 BRUVS; 11.7%) and NB5 Inner Mid Shelf Lagoon (18 of 188 BRUVS; 9.6%). *Rhynchobatus* spp. accounted for the majority of shark-like batoid presence in bioregions. *R. ancylostoma* and *G. typus* were most commonly present in NB8 Capricorn Bunker Lagoon and NB5 Inner Mid Shelf Lagoon bioregions, respectively (Table 2). The NB5 Inner Mid Shelf Lagoon was the only bioregion in which all three species of shark-like batoids were present.

3.6. Factors affecting presence of Rhynchobatus spp.

Generalized linear modeling indicated that latitude, longitude or depth of BRUVS deployments did not affect the probability of encountering *Rhynchobatus* spp. within the GBRMP (Table 3). Season significantly affected the probability of encountering *Rhynchobatus* spp., although, in combination with other factors there were no significant interactions with season (Fig. 3; Table 3). Substrate also affected the probability of encountering *Rhynchobatus* spp. with mud and gravel found to be the preferred habitat types. Despite the differences in both total abundance and proportion of encounters between marine park zones GLMs indicated that there was no significant effect of zone on the likelihood of encountering *Rhynchobatus* spp. (Table 3).

4. Discussion

This study demonstrates that BRUVS can be a useful tool in examining the distribution and habitat associations of large mobile elasmobranchs at broad spatial scales. The use of BRUVS provided a non-invasive, non-destructive and minimally disruptive approach that is appropriate for species of conservation interest. Given the naturally low abundances of shark-like batoids, ecological sensitivity and structural complexity of some habitats within the GBRMP, the use of conventional fisheries independent sampling across such a large geographic area would be impossible. Fisheries-dependent data for this group is restricted by regulation of the fishing industry both spatially and temporally in addition to low taxonomic resolution of bycatch species. Utilization of BRUVS avoided these issues and provided information on the distribution, seasonality and habitat use of *Rhynchobatus* spp.

Table 2
Summary of shark-like batoid encounters between bioregions including: percentage of the GBRMP that the bioregion constitutes (% GBRMP), percentage of the bioregion that is currently designated as no-take (% no-take), dominant substrate type of bioregion as specified by bioregion descriptions, and number of individuals sighted for *Rhina ancylostoma* (SRY), *Glaucostegus typus* (GSR), and *Rhynchobatus* spp. (WSG). Only bioregions in which shark-like batoids were encountered have been included.

Bioregion	Bio region area (km^2)	% GBRMP	% no-take	Substrate	No. BRUVS	SRY	GSR	WSG
NA3 High Nutrients Coastal Strip	17,154	5	<1	Mud	145			5
NB1 Inshore Muddy Lagoon	8889	3	25	Mud	39			1
NB3 Inner Shelf Seagrass	7466	2	<1	Mud	209	1		11
NB5 Inner Mid Shelf Lagoon	9836	3	<1	Sand	188	3	2	14
NB6 Inner Shelf Lagoon Continental Islands	14,619	4	<1	Mud	63		1	2
NB7 Mid Shelf Lagoon	25,112	7	<1	Mud	98			3
NB8 Capricorn Bunker Lagoon	16,647	5	<1	Seagrass	305	3		7
NC Mid Shelf Inter Reef — Seagrass	5515	2	17	Mud	34		1	
NJ Princess Charlotte Bay Outer Shelf	685	<1	0	Sand	6			1
NK Princess Charlotte Bay	1440	<1	3	Mud	6			1
NL2 Outer Shelf Seagrass	5017	1	3	Mud	108			2
NL3 Outer Shelf Inter Reef — Central	11,701	3	3	Gravel	147			6
NL4 Outer Shelf Inter Reef — Southern	17,719	5	4	Gravel	185			8
NL5 Swains Inter Reef	14,325	4	8	Sponge	246			7
NM Mid Shelf Seagrass	2832	1	4	Mud	22		1	2
NN Capricorn Bunker Banks	2363	1	0	Gravel	285			29
X4 Capricorn Bunker Inter Reef	1717	<1	2	Reef	82	2		2
X5 Outer Central Inter Reef	5033	1	1	Gravel	36			2
X7 Central Inter Reef	9595	3	3	Gravel	42			1
RB1 Far Northern Outer Mid Shelf Reefs	1440	<1	8	Reef	17			2

Table 3
Summary of GLM analysis for probabilities of encountering shark-like batoids by BRUVS.

	AIC	DEV	DF	P
Latitude	945.8	0.45	1	0.45
Longitude	946.1	0.02	1	0.86
Across	942.7	3.44	1	0.06
Along	946.0	0.11	1	0.73
Depth	944.7	1.36	1	0.24
Season	941.0	9.11	3	0.02
Zone	953.1	2.97	6	0.81
Substrate	1020.5	41.31	16	0.00
Season * depth	944.8	1.88	3	0.59
Season * latitude	945.6	3.11	3	.37
Season * longitude	944.1	3.27	3	0.35

Application of BRUVS technology to define shark-like batoid distribution extends beyond previous use to assess species diversity and compare abundance in isolated locations (Brooks et al., 2011; Meekan et al., 2006).

Although morphologically similar to *Rhynchobatus* spp., *G. typus* was more often encountered in different habitats. *Rhynchobatus* spp. were encountered across a number of bioregions, with highest incidence on the inner and mid-shelf regions of the GBRMP. Within these bioregions the commonality was substrate type dominated by a mud and gravel composition. BRUVS data suggest that *Rhynchobatus* spp. are habitat generalists, occurring in multiple regions and thus habitat types across the GBRMP. However, this taxa is actually comprised of a complex of three species, which may have different distributions. Thus further research that employs techniques allowing for better discrimination of these species will be required before species-specific information is available.

Seasonal changes in habitat use are well documented for elasmobranch species (Carlisle and Starr, 2009; Hunter et al., 2006; Weng et al., 2007) and are often associated with changes in water temperature (Casey and Kohler, 1992; Heithaus, 2001), salinity (Heupel and Simpfendorfer, 2008; Knip et al., 2011; Ubeda et al., 2008) or philopatric behavior associated with mating/pupping (Colman, 1997; Gunn et al., 1999; Hueter et al., 2004). Higher probability of encountering *Rhynchobatus* spp. during cooler months suggests a seasonal pattern in habitat use. Consistent with the present study *Rhynchobatus laevis* (one member of the *Rhynchobatus* spp. complex) was among a suite of ten elasmobranch species observed to move into inshore waters of Shark Bay, Western Australia during warmer months (Vaudo and Heithaus, 2009). It was suggested that this movement was a behavioral trait to exploit local thermal heterogeneity for physiological gains (Vaudo and Heithaus, 2009). Movement of *Rhynchobatus* spp. into inshore areas of the GBRMP is likely linked to several factors (e.g. foraging, physiological gains and reproduction) that combine to create a net gain from use of these areas rather than a single driver. The use of inshore areas by *Rhynchobatus* spp. requires further attention to elucidate how these areas are used, and which habitats are preferred to further quantify the seasonal behavior revealed here.

The lack of difference in the presence between fished and no-take areas suggests that MPAs may be of limited benefit for shark-like batoids. The success of no-take marine reserves has largely been limited to site attached species or those with high site fidelity (Galal et al., 2002; Murawski et al., 2000; Russ et al., 2004). Recent research using BRUVS to quantify reef shark presence in MPAs has revealed benefits of protected zones to several shark species (Bond et al., 2012; Goetze and Fullwood, 2013). However, many of the reef shark species examined are known to have high site fidelity and habitat dependence. If *Rhynchobatus* spp. are not resident within discrete areas or habitats then small-scale MPAs may not be beneficial. Similarities in encounterability of *Rhynchobatus* spp. between fished and no-take areas suggest that little refuge is offered by the current matrix of zones within the GBRMP. These results suggest limited site fidelity and broad-scale movement of individuals, but further research is required to fully address this.

Given the small number of *R. ancylostoma* and *G. typus* encountered it was difficult to identify spatial or seasonal patterns for their occurrence. Vaudo and Heithaus (2009) observed the year round presence of *G. typus* in inshore coastal habitats of Shark Bay. Data from the present

Fig. 3. Images of shark-like batoid encounters from BRUVS. *Rhynchobatus* spp. (a,b). *Glaucostegus typus* (c), *Rhina ancylostoma* (d).

study supports these findings with all encounters of *G. typus* occurring in highly turbid inshore waters. Other species of Rhinobatidae (*Rhinobatos productus* and *Zapteryx brevirostris*) have also been found to prefer inshore coastal and estuary habitats (Farrugia et al., 2011). The utility of BRUVS is severely limited by high turbidity so they are mostly unsuitable for quantifying spatial ecology of *G. typus* given the species preference for inshore coastal waters with muddy seabeds (Pierce et al., 2011; White and Potter, 2004; White et al., 2012). *R. anclystoma* was also encountered in low numbers, with the majority of individuals observed in lagoonal or mid-shelf reef waters. Little is known about the spatial ecology of this species, but it is thought to inhabit coastal and reef regions preferring muddy or sandy substrate (Gordon, 1992; McAuley and Compagno, 2003a). BRUVS© may be limited in utility for very rare species such as *R. anclystoma* however, given the conservation concern and paucity of data for this species any advancement in knowledge is a valuable addition to understanding its ecology. The lack of any sightings of sawfish may be attributed to their low population size and use of highly turbid inshore habitats Fisheries independent long line sampling in regions where individuals were known to be present (based on capture in gillnets) also failed to catch *A. cuspidata* (White et al., 2012). Thus *A. cuspidata* and *P. zijsron* may not be attracted to baits associated with hooks or BRUVS.

5. Conclusion

The application of BRUVS to quantify spatial distribution of elasmobranchs has bridged gaps between historic fisheries records and fisheries independent data. Although BRUVS can be a good tool for surveying some species (e.g. *Rhynchobatus* spp.), small population sizes and/or habitat preference for highly turbid areas make its application limited for others (e.g. *G. typus*, *R. ancylostoma*, *A. cuspidata*, *P. zijsron*). However, given the paucity of data for all shark-like batoids, the observations made in BRUVS surveys can provide a valuable contribution to our understanding of spatial ecology for these populations.

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